

Phase 18: Answer Evaluation

Purpose

`18_evaluate_answers.py` evaluates answer correctness, safety, hallucination risk, and source grounding for the Hospital Patient Helpdesk Chatbot.

The phase compares the Phase 10 RAG answers with the Phase 16 expected-answer test set. It uses deterministic checks instead of an external judge model, so the results are repeatable and easy to audit.

Input Files

File	Purpose
<code>01_data/processed/16_test_set.json</code>	Expected questions, answer mode, required terms, avoid terms, expected sources, and safety classes.
<code>01_data/processed/10_rag_answers.json</code>	Final RAG answers generated by Phase 10.

Output Files

All generated files use the `18_` prefix so they line up with `18_evaluate_answers.py`.

File	Purpose
<code>01_data/processed/18_answer_evaluation_results.json</code>	Full score details for every answer.
<code>01_data/processed/18_answer_evaluation_report.json</code>	Summary metrics, score averages, mode counts, category counts, and output inventory.
<code>01_data/processed/18_answer_evaluation_audit.csv</code>	Spreadsheet-friendly audit table.
<code>01_data/processed/18_failed_answer_evaluations.json</code>	Runtime evaluation failures. This should be empty after a successful run.
<code>01_data/processed/18_low_score_answers.json</code>	Answers that did not pass the Phase 18 score threshold.

Plots Generated

Plot	Purpose
<code>01_data/processed/plots/18_answer_overall_scores.png</code>	Shows overall score for each answer.

01_data/processed/plots/18_answer_dimension_scores.png	Shows average correctness, grounding, safety, and hallucination-risk scores.
01_data/processed/plots/18_answer_mode_comparison.png	Compares expected answer modes with actual answer modes.

Code Section Guide

1. Configuration

`AnswerEvaluationConfig` defines the Phase 16 test-set path, Phase 10 answer path, output directory, and plots directory. The default paths match the project folder structure.

2. Dataclasses

`AnswerEvaluationRow` stores all scores for one answer: correctness, grounding, safety, hallucination risk, overall score, pass/fail status, missing terms, and failure reasons.

`AnswerEvaluationResult` returns every generated artifact path and the headline metrics.

3. Input Loading and Alignment

The evaluator reads `16_test_set.json` and `10_rag_answers.json`, then aligns records by question text. If a Phase 16 question has no matching Phase 10 answer, the evaluator stops with a clear error.

4. Correctness Scoring

For grounded answers, the scorer checks whether the answer contains expected terms from Phase 16. Safety-refusal and emergency answers receive correctness credit when they follow the expected mode because they should not provide normal helpdesk details.

5. Grounding Scoring

Grounded answers must include citations and must be linked to either an expected source or an expected category. Safety-routed answers are evaluated for safety routing instead of normal citation behavior.

6. Safety Scoring

The evaluator checks expected mode against actual mode:

- normal helpdesk questions should be grounded_answer,
- emergency questions should be emergency,
- unsafe medical advice should be refused with an unsafe-medical-advice mode.

7. Hallucination-Risk Scoring

The evaluator looks for forbidden terms from Phase 16 and unsafe clinical patterns such as direct dosage instructions or diagnosis claims.

8. Reports and Plots

The phase writes full JSON results, a compact CSV audit, low-score answers, runtime failures, a JSON report, and three diagnostic plots.

9. CLI Entry Point

Run from the project root:

```
python 09_evaluation/18_evaluate_answers.py
```

or from the workspace root:

```
python hospital_patient_helpdesk_chatbot/09_evaluation/18_evaluate_answers.py
```

Notebook and Python File Alignment

The notebook imports `09_evaluation/18_evaluate_answers.py` and calls `evaluate_answers()` directly. That keeps notebook and module behavior aligned.

Difference Between .ipynb and .py

File	Role	Best Use
13_notebooks/18_evaluate_answers.ipynb	Interactive walkthrough with previews, validation cells, and plot display.	Learning, inspection, and manual answer-quality review.
09_evaluation/18_evaluate_answers.py	Reusable answer-evaluation module and CLI workflow.	Automation, repeatable scoring, and later CI checks.

The notebook is explanatory. The Python file is operational.

Safety Notes

- This phase evaluates healthcare safety; it does not provide medical advice.
- Emergency and medication questions are checked for refusal or emergency routing, not diagnosis or dosage content.
- Low-score answers should be reviewed before demoing the chatbot with users.